

DEYANG HSU

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EDUCATION

University of Southern California <i>Master's in Computer Science, Viterbi School of Engineering</i>	Los Angeles, California August 2024-May 2026 GPA: 3.875/4.0
The Ohio State University <i>Bachelor of Science in Electrical Computer Engineering</i> Honors: Magna Cum Laude Dean's List Scholar Alumni Society Scholarship recipient	Columbus, Ohio August 2018-May 2022 GPA: 3.769/4.0

Relevant Coursework:

Machine Learning & AI | OOP | Discrete Structures | Data Structures and Algorithms |Computer Architecture | Parallel Computing | Database systems | Operating Systems | Web Crawling and Search Engines | LLM | NLP | Cybersecurity | Computer Networking | Multimodal Probabilistic Learning | Natural Language Dialogue Systems

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL, HTML
Technologies & Tools: Text-to-LoRA, Doc-to-LoRA, TensorFlow, PyTorch, scikit-learn, XGBoost, LightGBM, Pandas, NumPy, HuggingFace, NLTK, OpenMP, CUDA, SLURM, Multithreading, MPI, Relational Databases, Web Crawling, RAG
Systems: RISC-V, CISC, GitHub, Subversion, Networking, Operating Systems, Computer Architecture

RESEARCH EXPERIENCE

Research Assistant, FORTIS Lab University of Southern California	Jan. 2025 – May 2026
- Architected a spatio-temporal wildfire prediction pipeline fusing ERA5 weather reanalysis and LANDFIRE vegetation indices (LAI, NDVI) across 116,000+ time-county observations under Professor Yue Zhao. - Designed a Hybrid Mamba (SSM) and Temporal Fusion Transformer with GCN spatial message passing for long-range spatio-temporal dependencies, achieving a 15% F1 gain; engineered Hybrid Poisson/BCE loss to address extreme class imbalance. - Conducted CNA research leveraging frontier LLMs (GPT-4, DeepSeek, Claude) for multi-narrative evaluation and conflict detection.	

PUBLICATIONS

[The Traffickers' Pitch: Detecting Deceptive Recruitment in Online Job Boards](#)
S. Zhou, P. Qiu, T. Salkar, L.B. Urrutia, D. Shen, **D. Hsu**, E.C. Choi, E. Ferrara. *arXiv preprint arXiv:2605.25416*, 2026

PROJECTS

StressID: Physiological Audio Stress Detection Pipeline
- Designed and built an end-to-end audio ML pipeline for real-time stress detection from physiological audio signals, combining signal processing with deep learning classification models. - Implemented acoustic feature extraction modules (MFCCs, pitch, prosodic features, voice activity detection), enabling precise discrimination of stress states from audio biomarkers. - Fine-tuned pretrained speech models using LoRA adapters via HuggingFace, adapting large audio transformers for stress recognition with minimal computational overhead.

WORK EXPERIENCE

Intel Software Development Engineer	Chandler, Arizona June 2022-August 2023
- Diagnosed and resolved critical bugs across in-house software services, driving a 27% increase in system uptime and reducing production incidents - Refactored production database systems with PL/SQL, improving query performance, data reliability, and throughput - Developed and deployed custom production software solutions, leading to an 18% rise in wafer production efficiency	
General Motors Manufacturing Controls Engineer Intern	Bay City, Michigan June 2021-August 2021
- Diagnosed and resolved failures in robotics systems and PLCs, cutting production downtime by 8% - Implemented an IO-Link pressure sensor system via Siemens SIMATIC STEP 7, reducing inventory scrapping by 13%	